First Impressions under Scarcity: An Economic Conjoint Experiment for Studying Electoral Judgment

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Abstract

How do voters decide whether first impressions are sufficient, or whether additional information is worth acquiring before making a political judgment? Research in political psychology shows that facial appearance and other low-information cues shape candidate evaluations quickly and powerfully, while the conjoint literature typically assumes that all profile attributes are fully revealed from the outset. This paper introduces a new method—an economic conjoint experiment—that bridges these traditions by making information acquisition endogenous and costly. Respondents first encounter real candidate photographs in a low-information setting and then decide whether to spend points to unlock additional profile attributes before making a judgment. In some tasks, time pressure is randomized, allowing the design to identify how costly deliberation changes political evaluation. The method yields behavioral quantities of interest that standard fully revealed conjoint designs are not built to capture. In particular, I introduce *marginal information demand* (MID), an estimand that summarizes the probability that a given piece of information is acquired under specified profile and task conditions. Substantively, the paper applies this design to the study of first impressions and gendered scrutiny in electoral judgment, arguing that female candidates may elicit greater information demand because they are evaluated under a higher burden of verification.

Introduction

Research in political psychology and political behavior has long shown that voters rely on low-information cues when evaluating candidates (Lau and Redlawsk 2001), especially facial appearance (D. Dion 1998; Stockemer and Praino 2019a), perceived competence (Herrmann and Shikano 2016), attractiveness (Rosar, Klein, and Beckers 2008; K. Dion, Berscheid, and Walster 1972), trustworthiness, and social-class signals (Bahamonde and Sarpila 2024). At the same time, work in behavioral decision-making suggests that information use is shaped not only by limited cognition (Redlawsk 2001) but also by the costs of acquiring and processing additional evidence (Zaller 1992). Yet these literatures have rarely been integrated in a way that treats information search itself as part of the political judgment process (but see Lau and Redlawsk 2006). Most existing studies ask whether appearance affects evaluations (Berggren, Jordahl, and Poutvaara 2010; Berggren, Jordahl, and Poutvaara 2017), or whether providing new information changes preferences (Todorov et al. 2005; Mattes et al. 2010). While prior research has examined how voters *search* among available information (Lau and Redlawsk 2006), much less is known about how they behave when further information must be *acquired under explicit budget constraints*, and whether those *costly search decisions* are themselves structured by gender.

This gap is especially important because first impressions appear to emerge extremely quickly and to persist even when additional information is available. The literature on candidate evaluation consistently suggests that photographs allow voters to form immediate impressions about candidates (Banducci et al. 2008). Related work shows that such impressions can arise under extremely short exposure times (e.g., 100 milliseconds) and are often only weakly corrected by subsequent information (Todorov et al. 2005; Mattes et al. 2010). Antonakis and Dalgas (2009) even report that children have the same (and sometimes even *better*) predicting skills when choosing winning political candidates. **If so, the key question is not simply whether facial appearance matters, but under what conditions respondents *choose* to verify, override, specially in context of resource scarcity**.

This paper argues that electoral judgment is not only a matter of which candidate traits voters prefer, but also of how much resources (cognitive and monetary) they are willing to expend to learn about different candidates. That question is especially important in the case of women in politics. A large literature suggests that female candidates are often evaluated through gendered expectations and may face distinct standards of competence, likability, and suitability for office. But the existing evidence focuses mainly on vote choice (Bahamonde and Sarpila 2024), candidate evaluations, or stereotype activation (Ditonto and Mattes 2018; Krook and Restrepo Sanín 2020). It tells us whether gender matters for outcomes, but not whether gender *shapes the process of acquiring information itself.* We contend that one of the key mechanisms of unequal political judgment may lie precisely there: **female candidates may elicit greater information demand because respondents feel a stronger need to verify, resolve, or reconcile their initial impressions**. In that sense, political judgment is distributed unevenly across candidates.

Our core theoretical claim is that first impressions generate unequal demand for additional information. When respondents believe that a candidate's face already provides a sufficiently clear signal, they may rationally decline to spend scarce resources on further investigation. Low search may therefore reflect efficient economizing if first impressions and facial heuristics are informative. The gendered implication is especially important: if female candidates are judged under stricter priors (Ditonto and Mattes 2018; Krook and Restrepo Sanín 2020), respondents may be less willing to trust first impressions alone and more likely to invest in additional search. The result is a stronger demand for information not because women inherently provide less information, but because initial impressions of female candidates may activate motivated reasoning (Kunda 1990), inducing respondents to seek and evaluate additional evidence under a higher burden of verification.

To study this process, we introduce a novel *economic conjoint experiment* implemented in oTree (see prototype here). The design combines the multidimensional choice structure of conjoint analysis (Hainmueller, Hopkins, and Yamamoto 2014) with an explicitly economic search environment in which information is scarce, costly, and *endogenously* acquired: rather than presenting respondents with a fully revealed bundle of candidate attributes, we allow them to uncover information selectively by spending points from a limited budget. This transforms the conjoint from a *passive* stated-preference task into a sequential decision problem in which *search behavior* itself becomes theoretically meaningful. Following Antonakis and Dalgas (2009), respondents first encounter real candidate photographs in a low-information setting, form an initial judgment, but then decide whether to pay to reveal additional profile attributes before updating their choice (see Figure 1). The experiment therefore identifies not only which candidate features affect electoral judgments, but also which features respondents consider worth learning about, under what conditions, and for whom.

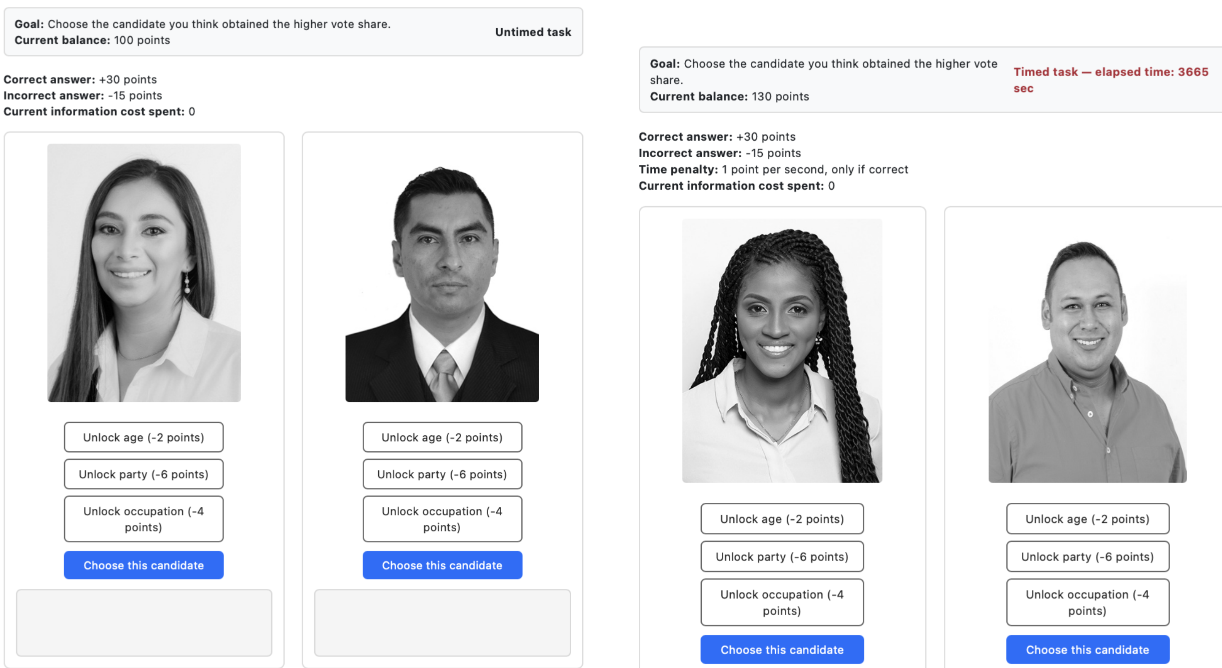
To capture this acquisition stage directly, we introduce a new estimand, which we call *marginal information demand* (MID), for attribute m:

$$MID_m(z, s) = \Pr(U_{ijtm} = 1 \mid Z_{ijt} = z, S_{it} = s),$$

where $MID_m(z, s)$ denotes the probability that attribute m is unlocked under profile condition z and task condition s; $U_{ijtm} = 1$ indicates that respondent i unlocks attribute m for profile j in task t; Z_{ijt} denotes the profile-level condition of substantive interest, such as candidate gender in the present application; and S_{it} denotes the relevant task condition. In the present design, one such task condition is time pressure. In some choice tasks assigned at random, respondents face a visible timer and a reward structure in which delay reduces the value of a correct answer. The purpose of this feature is to isolate the conditions under which respondents rely more heavily on first impressions when deliberation becomes costly. If rapid perceptual inferences are indeed powerful and persistent, then timed tasks provide a direct way to examine whether respondents fall back on facial heuristics when the opportunity to reflect is constrained.

This methodological contribution is substantial. Standard conjoint experiments are powerful tools for estimating the causal effects of profile attributes on selection probabilities (Hainmueller, Hopkins, and Yamamoto 2014; Leeper, Hobolt, and Tilley 2020), but they typically reveal much more about respondents' *final* choices than about the information-processing strategies that produced them. Moreover, recent work using eye-tracking shows that respondents in conjoint experiments selectively attend to *some* attributes and profiles while ignoring others, in ways consistent with bounded rationality and cognitive cost minimization (Jenke et al. 2021). Based on the dynamic process tracing environment (DPTE) developed by Lau and Redlawsk (2006), our economic conjoint experiment extends these insights by making information acquisition itself endogenous but also *costly*. **Rather than inferring information use from researcher-imposed, fully revealed conjoint profiles, we design an environment in which respondents must decide for themselves whether additional information is worth acquiring before making a decision**.

We expect respondents to spend more points when visual cues are ambiguous or incongruent. We also expect this logic to be gendered: **female candidates should trigger greater information demand**, consistent with a politics of asymmetric scrutiny in which women are evaluated under a higher burden of verification.



Untimed task | Timed task

Figure 1: Prototype screens of the economic conjoint experiment

Technical Methodological Addendum

Core design features

- **Unit of evaluation.** Each respondent $i = 1, \dots, N$ completes T repeated choice tasks. In each task $t = 1, \dots, T$, the respondent sees two candidate profiles, indexed by $j \in \{L, R\}$.
- **Endogenous information acquisition.** For each profile, respondents can selectively unlock three attributes: party, age, and occupation (see **Bahamonde2024**). Let $m \in \{\text{party}, \text{age}, \text{occupation}\}$ index the attribute. Unlocking is costly and therefore behaviorally meaningful.
- **Randomization of candidate exposure.** The two profiles shown in each task, indexed by $j \in \{L, R\}$, are randomly drawn from the candidate pool and then randomly assigned to the left or right side of the screen. This randomization ensures that any side-specific effect is orthogonal to candidate characteristics.
- **Randomization of information order.** The vertical order of the three unlockable attributes $\{\text{party}, \text{age}, \text{occupation}\}$ is randomized at the respondent level and then held fixed across tasks $t = 1, \dots, T$ for that respondent. This prevents systematic ordering effects while preserving within-respondent comparability.
- **Randomization of time pressure.** Time pressure is randomized at the task level. A given task may or may not include the timer, so respondents can experience untimed tasks only, timed tasks only, or a mixture of both across the survey.
- **Budget-constrained search.** Each respondent receives an initial endowment and must manage a running point balance across tasks. Correct choices increase the balance, incorrect choices reduce it, and each information unlock carries an explicit cost. In timed tasks, delay further reduces the net payoff from a correct choice.
- **Behavioral measurement.** The design records which attributes are unlocked, in which task, for which profile, how much is spent on information, how much time is used, whether the respondent was correct, and the respondent's balance after each task.
- **Substantive outcome.** In the current implementation, respondents are asked to identify which candidate obtained the higher vote share. This aligns the forced choice with an externally observable electoral outcome rather than with an unobserved personal preference.

Main quantities of interest

The central advantage of the design is that it yields causal quantities not only for final choices, but also for the *search process* itself. Let Z_{ijt} denote the profile-level condition of substantive interest, such as candidate gender in the present application, and let $S_{it} \in \{0, 1\}$ denote whether task t is timed.

Let

$$U_{ijtm} \in \{0, 1\}$$

indicate whether respondent i unlocks attribute m for profile j in task t.

To emphasize that the key behavioral object is *information demand*, define the *marginal information demand* (MID) for attribute m as

$$MID_m(z, s) = \Pr(U_{ijtm} = 1 \mid Z_{ijt} = z, S_{it} = s).$$

MID is therefore the probability that attribute m is unlocked under profile condition z and task condition s.

The first quantity of interest is the effect of the profile-level condition on the probability that a given attribute is unlocked:

$$\tau_m = MID_m(1,0) - MID_m(0,0).$$

In the present application, τ_m captures whether female candidates generate greater demand for information on attribute m in the absence of time pressure. If the theory of asymmetric scrutiny is correct, we expect $\tau_m > 0$ for at least some attributes.

A second quantity of interest is the effect of time pressure on information acquisition:

$$\delta_m = MID_m(0,1) - MID_m(0,0).$$

This estimand captures whether respondents become less likely to acquire costly information when decision time is made more salient.

The key interaction quantity of interest is:

$$\gamma_m = [MID_m(1,1) - MID_m(0,1)] - [MID_m(1,0) - MID_m(0,0)].$$

This difference-in-differences estimand captures whether the gap in information demand associated with the profile-level condition changes under time pressure. In the present application, it is the most direct formalization of the claim that female candidates may face a higher burden of verification, especially when respondents are forced to make decisions under scarcity.

Because the design also records the total amount spent on information, a second family of quantities of interest concerns search intensity. Let

$$C_m > 0$$

denote the point cost of unlocking attribute m. Define total information expenditure for respondent i in task t as

$$E_{it} = \sum_{j \in \{L,R\}} \sum_m C_m U_{ijtm}.$$

Then the average causal effect of the profile-level condition on total search expenditure can be written as

$$\tau_E = \mathbb{E}[E_{it}(1,0) - E_{it}(0,0)],$$

with the obvious timed analogue and interaction extension.

The design also permits causal analysis of final task performance. Let

$$Y_{it} \in \{0,1\}$$

indicate whether respondent i correctly identifies the candidate with the higher vote share in task t. Then one can define:

$$\tau_Y = \mathbb{E}[Y_{it}(1,0) - Y_{it}(0,0)],$$

to study whether the profile-level condition affects prediction accuracy, and

$$\gamma_Y = \mathbb{E}\big[(Y_{it}(1,1) - Y_{it}(0,1)) - (Y_{it}(1,0) - Y_{it}(0,0))\big],$$

to assess whether time pressure differentially changes accuracy across profile conditions.

Number of tasks

For the present study, I recommend using

$$T = 8$$

choice tasks per respondent.

This choice is methodologically conservative. It sits comfortably within the logic of Bansak et al. (2018), who find little evidence of substantial deterioration in conjoint responses even with much larger numbers of tasks. Using eight tasks therefore provides a reasonable compromise between statistical power and respondent burden, especially since the present design is cognitively more demanding than a standard fully revealed conjoint because respondents must actively manage costly information acquisition.

Sample size

The main power calculation should be anchored in the candidate-level information acquisition outcome U_{ijtm}, since the proposal's central theoretical claim concerns differential information demand. Suppose the target effect size is a five-percentage-point difference in the probability of unlocking a given attribute in untimed tasks, for example:

$$MID_m(1,0) - MID_m(0,0) = 0.05.$$

Using a conventional two-sided test with $\alpha = 0.05$ and power 0.80, and taking a baseline unlock probability of roughly 0.30, the required independent sample size is approximately

$$n_{\text{indep}} \approx 1{,}376$$

profile observations per arm.

In the present design, however, observations are clustered within respondents because each respondent evaluates multiple profiles. With $T = 8$ tasks and two profiles per task, each respondent contributes up to

$$m = 2T = 16$$

candidate-profile exposures. A standard design-effect correction gives:

$$DE = 1 + (m-1)\rho,$$

where ρ is the within-respondent intraclass correlation in profile-level outcomes.

For plausible values such as $\rho \in [0.10, 0.20]$, this yields:

$$DE \in [2.5, 4.0].$$

The effective number of profile exposures per arm is therefore approximately

$$n_{\text{eff, arm}} \approx \frac{8N}{DE},$$

because, under balanced randomization, each respondent contributes on average about $T = 8$ profile exposures to each arm of the profile-level condition.

Setting

$$\frac{8N}{DE} \geq 1{,}376$$

implies:

$$N \geq \frac{1{,}376 \times DE}{8}.$$

This gives a required respondent sample approximately in the range:

$$N \in [430, 688].$$

That range is appropriate for the *main effect* of the profile-level condition on attribute unlocking. Because the project is also interested in interactions with time pressure and in heterogeneity across multiple unlockable attributes, a more conservative target is warranted. I therefore recommend a target sample of:

$$N = 1{,}000$$

respondents, with

$$T = 8$$

tasks each.

This design would generate:

$$2NT = 16{,}000$$

candidate-profile exposures in total. Under balanced assignment, this yields substantial information for estimating both main effects and interactions with time pressure while remaining comfortably within the range of task counts that Bansak et al. (2018) suggest are unlikely to create severe satisficing problems.

Relation to AMCEs, Marginal Means, and Framing Designs

It is useful to distinguish clearly between the quantities of interest that dominate the standard conjoint framework and the quantities of interest targeted by the present design. In the Hainmueller, Hopkins, and Yamamoto (2014) framework, the principal causal estimand is the average marginal component effect (AMCE), which summarizes the effect of changing one attribute level on the probability that a profile is chosen, averaging over the joint distribution of the remaining attributes. In their notation, if $Y_{ijt}(x)$ denotes the potential choice outcome for respondent i in task t when profile vector x is shown, and if $X_{ijt\ell}$ denotes the ℓth attribute of the profile, then the AMCE for attribute ℓ compares two levels, say x_1 and x_0, averaging over all possible realizations of the remaining profile components. Substantively, the AMCE answers the question: by how much does support for a profile change when one attribute is shifted from x_0 to x_1, holding fixed the fully revealed conjoint structure and averaging over the distribution of the other attributes?

Leeper, Hobolt, and Tilley (2020) show that AMCEs are closely tied to marginal means. In a fully randomized forced-choice conjoint, a marginal mean for feature level x can be written as

$$MM_\ell(x) = \Pr(Y_{ijt} = 1 \mid X_{ijt\ell} = x),$$

where the probability is taken over the randomized distribution of all remaining profile attributes. In fully randomized designs, the AMCE is simply the difference between two such marginal means:

$$AMCE_\ell(x_1, x_0) = MM_\ell(x_1) - MM_\ell(x_0).$$

Thus, AMCEs summarize relative differences against a reference level, whereas marginal means summarize the absolute favorability of profiles carrying a given feature level. In both cases, however, the estimands are defined over *fully revealed* profile content and linked to the respondent's final choice.

The present design differs because the central behavioral question arises *before* the final choice is made. Respondents do not receive the full profile ex ante. Instead, they first observe the candidate's photograph and then decide whether to unlock party, age, and occupation (among other attributes) under a binding budget and, in some tasks, under time pressure. As a result, the primary quantities of interest are not initially AMCEs of fully displayed attributes on the final choice outcome Y_{it}, but quantities governing *information acquisition itself*. Let

$$U_{ijtm} \in \{0, 1\}$$

indicate whether respondent i unlocks attribute $m \in \{\text{party}, \text{age}, \text{occupation}\}$ for profile j in task t. Let Z_{ijt} denote the profile-level condition of substantive interest and let $S_{it} \in \{0, 1\}$ indicate whether task t is timed.

To emphasize that the key behavioral object is *information demand*, it is useful to define the *marginal information demand* (MID) for attribute m as

$$MID_m(z, s) = \Pr(U_{ijtm} = 1 \mid Z_{ijt} = z, S_{it} = s),$$

that is, the probability that attribute m is unlocked under profile condition z and task condition s. In direct analogy to the way marginal means summarize support levels in standard conjoint analysis, MID summarizes the level of demand for a given piece of information before the final choice is made. The corresponding contrast on the profile-level condition can then be written as

$$\tau_m = MID_m(1, 0) - MID_m(0, 0),$$

and the time-pressure interaction can be expressed as

$$\gamma_m = \big[MID_m(1, 1) - MID_m(0, 1)\big] - \big[MID_m(1, 0) - MID_m(0, 0)\big].$$

Table 1: Benchmarking standard conjoint estimands against the economic conjoint acquisition estimand

Estimand	**Notation**	**What it measures**	**Stage of decision process**
AMCE (Hainmueller, Hopkins, and Yamamoto 2014)	$AMCE_\ell(x_1, x_0) = MM_\ell(x_1) - MM_\ell(x_0)$	Average causal effect of changing attribute ℓ from x_0 to x_1 on profile choice, averaging over the remaining fully revealed attributes	Final choice given full exposure
Marginal Mean (Leeper, Hobolt, and Tilley 2020)	$MM_\ell(x) = \Pr(Y = 1 \mid X_\ell = x)$	Probability that a fully revealed profile with attribute level x is chosen	Final choice given full exposure
Marginal Information Demand (MID)	$MID_m(z, s) = \Pr(U_{ijtm} = 1 \mid Z_{ijt} = z, S_{it} = s)$	Probability that attribute m is unlocked under profile condition z and task condition s	Information acquisition before final choice

Notes: Y denotes profile choice, X_ℓ a fully revealed attribute, and U_{ijtm} the decision to unlock attribute m for profile j shown to respondent i in task t. AMCEs and marginal means are standard estimands from fully revealed conjoint analysis. MID is the analogous level quantity for the acquisition stage of the present design.

These estimands do not contradict AMCEs or marginal means. Rather, they operate at a different stage of the decision process (see Table 1). AMCEs and marginal means ask: *once an attribute is part of the revealed profile, how does it shape support?* MID instead asks: *is that attribute acquired at all, under what conditions, and for whom?* In this sense, the current design *complements* the standard conjoint framework by shifting attention from the causal effects of fully revealed attributes on final choice to the causal structure of selective information acquisition under scarcity. That is precisely the dimension that standard fully revealed conjoint designs are not built to capture.

This distinction also clarifies the limits of standard AMCE and marginal-mean analyses for the present application. If one were to compute an AMCE of party, age, or occupation on final choice in the usual way, one would be analyzing the consequences of those attributes *conditional on exposure*. But in the present design, exposure is itself endogenous: respondents must choose whether to incur a cost in order to observe the attribute. For that reason, standard AMCEs and marginal means cannot by themselves recover the key behavioral object of interest, namely the process by which some pieces of information become visible while others remain unseen. At most, they can describe how information matters *after it has entered the respondent's choice set*. The present design is therefore best understood as extending, rather than replacing, the conjoint framework: it preserves the multidimensional profile-comparison logic of conjoint analysis while introducing a prior stage in which information acquisition itself becomes a theoretically meaningful outcome. In that extended framework, MID plays a role analogous to the marginal mean, but for the acquisition stage rather than the choice stage.

It is equally important to clarify that this is not a standard framing experiment. In a conventional framing design, respondents are typically assigned to one of a small number of externally imposed messages or interpretive presentations, and the central estimand compares outcomes across those frame assignments. Formally, if $F_i \in \{0, 1\}$ indexes exposure to two alternative frames and $Y_i(F_i)$ denotes the corresponding potential outcome, the canonical framing estimand is

$$\Delta_F = \mathbb{E}[Y_i(1) - Y_i(0)].$$

The frame is exogenous, uniform within condition, and delivered directly by the researcher. The inferential target is the causal effect of *how the same object is presented*. By contrast, the present design does not assign respondents to a single message frame that exogenously alters interpretation. Instead, it assigns respondents to a decision environment in which multiple profile components may or may not become visible depending on respondents' own acquisition choices. What varies is not simply the wording or presentation of a fixed object, but the respondent's self-directed exposure to profile information under cost and time constraints.

Using notation, a framing experiment takes the form

$$F_i \longrightarrow Y_i,$$

whereas a standard fully revealed conjoint takes the form

$$X_{ijt} \longrightarrow Y_{ijt},$$

with X_{ijt} denoting the vector of randomized profile attributes. The present design instead has the structure

$$\text{image}_{ijt} \longrightarrow U_{ijtm} \longrightarrow Y_{it},$$

possibly moderated by task-level time pressure S_{it}. The crucial difference is that the information set entering the final judgment is not wholly assigned by the researcher at the moment of treatment delivery. It is jointly produced by the experimental environment and the respondent's own acquisition decisions. For that reason, the design is not well described as a framing experiment, even though it manipulates the informational environment. Its distinctive feature is not exogenously imposed interpretation, but endogenous, costly, and sequential revelation of profile content.

Econometric specification

The empirical analysis of MID is conducted at the profile-attribute level. Let respondent $i = 1, \ldots, N$ complete tasks $t = 1, \ldots, T$, and let each task contain profiles $j = 1, \ldots, J_t$. In the current two-profile implementation, $J_t = 2$, but the notation allows the number of profiles per task to vary.

Let

$$\mathcal{M} = \{1, \ldots, M\}$$

denote the set of unlockable attributes. In the present application, these attributes include party, age, and occupation, but the econometric structure is more general: m indexes any piece of profile information that can be acquired at a cost before the final choice is made.

For each respondent, task, profile, and attribute, define

$$U_{ijtm} = \begin{cases} 1 & \text{if respondent } i \text{ unlocks attribute } m \text{ for profile } j \text{ in task } t, \\ 0 & \text{otherwise.} \end{cases}$$

The dependent variable in the main analysis is therefore not the realized value of the attribute itself, but the respondent's decision to acquire that attribute. For example, when m indexes age, the outcome is whether the respondent paid to learn the candidate's age; when m indexes occupation, the outcome is whether the respondent paid to learn the candidate's occupation. This distinction is central to the design. Attributes that are hidden until purchased are not initially observed components of a fully revealed conjoint profile. They are objects of endogenous information demand.

Let Z_{ijt} denote a profile-level condition that is visible or inferable before the acquisition decision is made. In the present application, Z_{ijt} is candidate gender as inferred from the photograph, but the same framework applies to any pre-acquisition profile condition, such as apparent age (Keating, Randall, and Kendrick 1999, p. 594), facial competence (Todorov et al. 2005; Herrmann and Shikano 2016), perceived social class (Bahamonde and Sarpila 2024), attractiveness (Banducci et al. 2008; Budesheim and DePaola 1994; K. Dion, Berscheid, and Walster 1972; Ditonto and Mattes 2018; Hart, Ottati, and Krumdick 2011; Kalick 1988; Praino, Stockemer, and Ratis 2014; Praino and Stockemer 2019; Stockemer and Praino 2017; Stockemer and Praino 2019b; Wigginton and Stockemer 2021), or other image-based cues (Price et al. 2011). Let $S_{it} \in \{0, 1\}$ indicate whether task t is conducted under time pressure. The basic attribute-specific linear probability model is

$$U_{ijtm} = \alpha_m + \beta_m Z_{ijt} + \lambda_m S_{it} + \gamma_m (Z_{ijt} \times S_{it}) + \delta_m L_{ijt} + f_m(t) + \varepsilon_{ijtm},$$

where L_{ijt} indicates the profile's screen position and $f_m(t)$ denotes task-order adjustments, such as a linear task number or task fixed effects. The coefficient β_m estimates the difference in the probability of unlocking attribute m associated with the profile-level condition in untimed tasks. In the gender application, β_m captures whether female candidates generate greater demand for attribute m than male candidates when respondents are not under time pressure. The coefficient λ_m captures the effect of time pressure for the baseline profile condition, while γ_m captures whether time pressure changes the profile-condition gap in information demand.

This specification is the regression analogue of the MID estimands defined above. In particular, when Z_{ijt} and S_{it} are binary,

$$\beta_m = MID_m(1, 0) - MID_m(0, 0),$$

and

$$\gamma_m = [MID_m(1, 1) - MID_m(0, 1)] - [MID_m(1, 0) - MID_m(0, 0)],$$

up to the included design controls. Thus, the model directly estimates the level and contrast quantities implied by the acquisition-stage estimand.

The main analyses estimate separate models by unlockable attribute. That is, the model is estimated once for each $m \in \mathcal{M}$. This produces a set of attribute-specific MID estimates:

$$\widehat{\beta}_1, \widehat{\beta}_2, \ldots, \widehat{\beta}_M,$$

and, when time pressure is included,

$$\widehat{\gamma}_1, \widehat{\gamma}_2, \ldots, \widehat{\gamma}_M.$$

This is preferable to focusing first on a pooled model because the substantive meaning of information demand may differ across attributes. A higher probability of unlocking party does not necessarily have the same theoretical interpretation as a higher probability of unlocking age, occupation, or any other profile attribute. Presenting separate-attribute MID estimates therefore preserves the behavioral interpretation of each acquisition decision.

Because the same respondent makes repeated acquisition decisions across tasks, profiles, and attributes, standard errors must account for within-respondent dependence. The baseline inference clusters standard errors at the respondent level. When the same profile images are shown to multiple respondents, a more conservative specification clusters standard errors by both respondent and candidate image. This accounts both for repeated decisions by the same respondent and for dependence induced by repeated exposure to the same candidate photograph across respondents.

The specification also clarifies how hidden attribute values should enter the analysis. The realized value of an unlockable attribute, such as the candidate's actual age or occupation, exists in the data before it is revealed to the respondent. However, because the respondent does not observe that value unless the attribute is purchased, it should not be treated as a direct cause of the unlocking decision unless it is visible or inferable from pre-acquisition information. For example, apparent age inferred from the photograph may affect whether respondents decide to unlock age, but the hidden numerical age itself cannot directly affect the acquisition decision except through visible cues or correlations known to the respondent. For this reason, the main MID models condition only on randomized task features, pre-acquisition profile conditions, screen position, and task order. The realized contents of unlocked attributes are analyzed only after acquisition, and are therefore not part of the primary MID specification.

The Choice Task

The experiment requires a choice task that is at once politically meaningful and incentive compatible. A direct vote-choice question would satisfy the first criterion, but it would sit uneasily with the logic of the design. The purpose of the experiment is not simply to estimate which candidate profile respondents prefer once all attributes are displayed. It is to study which pieces of candidate information respondents consider worth acquiring before making a political judgment. The task must therefore give respondents a reason to economize on information while also giving them a reason to seek information when they believe it will improve their decision.

This requirement rules out a purely expressive preference question. If respondents are asked which candidate they personally prefer, their answers cannot be treated as right or wrong. A respondent may prefer a candidate who lost, or reject a candidate who won, without making a mistake. Scoring such a response against the electoral outcome would implicitly treat the majority choice as the correct choice, or electoral success as evidence of candidate quality. The design avoids that assumption by asking respondents to make a prediction rather than to state a preference.

The respondent-facing task is therefore:

> *Imagine that residents in this municipality had to choose one of these two candidates to lead an important local project. Based on the information available to you, which candidate do you predict residents would choose?*

This formulation combines two features that are useful for the present design. The first is a hypothetical leadership frame. Rather than asking respondents to forecast an election in the abstract, the question places the candidates in a concrete municipal role. This follows the logic of experimental work showing that seemingly simple leadership-selection tasks can elicit politically relevant judgments about competence, suitability, and appeal (Antonakis and Dalgas 2009).

The second feature is an explicit prediction frame. Respondents are not asked which candidate they would choose, or which candidate deserves to be chosen. They are asked which candidate local residents would choose. This makes the response consequential in a defensible way. Because the profiles correspond to real municipal candidates, predictions can be scored against an external electoral benchmark. In this respect, the task follows election-prediction designs in which respondents make prospective judgments about uncertain political outcomes that can later be compared to realized results (Kara-Yakoubian and Spaniol 2025).

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